

AYMANE AIT DADS

Generative AI Intern | LLM Fine-Tuning · RAG Pipelines · Agentic AI Applications

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PROFESSIONAL SUMMARY

Ranked 17th out of 3,000+ teams (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a 30B-parameter model with LoRA SFT, staged continuation training, and task-level error analysis on 5 structured reasoning categories. Built and optimized end-to-end LLM fine-tuning pipelines using LoRA, PEFT, Unsloth, and Hugging Face Transformers, reducing hard-task failure rates through ablation-driven diagnosis that reasoning over-expansion - not wrong reasoning - was the primary failure mode. Brings hands-on experience with RAG pipelines, Claude API, LangChain, agentic pipelines, REST API integration, prompt engineering, and tool-calling to build production-grade generative AI applications for enterprise systems. Aims to design and deploy scalable LLM-powered copilots and document intelligence systems at Openkyber, applying fine-tuning and retrieval-augmented generation expertise to real-world enterprise challenges.

CORE COMPETENCIES

LLM Fine-Tuning (LoRA) | RAG Pipeline Development | Prompt Engineering | Agentic Pipelines | REST API Integration | Generative AI Applications | Document Intelligence | Vector Database Integration

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Engineered automated **feature engineering and model evaluation pipeline** using Python, Scikit-learn, and Pandas on 100,000+ fiber-optic network records, reducing manual analysis time by ~60%.
- Built a **Random Forest + K-Means clustering pipeline** classifying network quality across thousands of nodes by upload speed, latency, and jitter metrics.
- Developed a customer performance profiling system mapping outputs to **5 commercial service tiers**, adopted directly by the network optimization team for maintenance decisions.
- Delivered stakeholder presentations translating model outputs into **actionable maintenance recommendations**, communicating technical findings to non-technical business teams via Power BI dashboards.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context) across 5 structured reasoning categories.
- Performed **task-level error analysis** across 5 reasoning categories (unit conversion, cipher, symbolic, gravity, bit manipulation); built a diagnostic evaluation pipeline revealing that reasoning over-expansion - not incorrect reasoning - was the primary failure mode, and built custom cipher parsers and 8-bit operation solvers to produce verified chain-of-thought training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Kaggle GPU

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in EURECOM class** private Kaggle leaderboard (0.791 AUC) by fine-tuning CLIP ViT-L/14 (428M parameters) with a custom MLP head, dual learning rates, and 10-view TTA ensemble across 250,000 training images from 25+ generator types.
- Submitted to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; identified that 1-epoch training (0.793 AUC) outperforms 4-epoch training (0.767 AUC), demonstrating superior out-of-distribution generalization through controlled ablation experiments.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · AdamW

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Cloud Computing, Statistics, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

LLM Fine-Tuning & Generative AI: LoRA | QLoRA | SFTTrainer | Hugging Face Transformers | Unsloth | PEFT | vLLM | chain-of-thought | prompt engineering

AI Applications & RAG: RAG pipelines | LangChain | Claude API | agentic pipelines | tool-calling | REST APIs | Supabase | PostgreSQL | Docker

ML Engineering & Data Science: PyTorch | TensorFlow | Scikit-learn | Pandas | NumPy | BF16/FP16 mixed precision | ablation studies | Git | Linux